# CHEMISTRY

**Time: 3 Hours** **Total Marks: 70**

**Note:**

i. All questions are compulsory.
ii. Answers of both the sections should be written in same answer book.
iii. Draw well labelled diagrams and write balanced equations wherever necessary.
iv. Figures to the right indicate full marks.
v. Use of logarithmic table is allowed.
vi. Every new question must be started on a new page.

## SECTION – I

**Q.1. Attempt any SIX:** **[12]**

i. What is the role of the following compounds-
   a. $CaF_2$ in metallurgy of aluminium?
   b. $SiO_2$ in the extraction of copper from copper pyrites?

ii. Classify the following molecular solids into different types:
   a. HCl b. $CO_2$
   c. Solid ice d. $SO_2$

iii. Define the following terms:
   a. Cryoscopic constant b. Resistivity

iv. Arrange the following oxyacids of chlorine in decreasing order of their thermal stability. Give reason.
   $HOClO$, $HOCl$, $HClO_4$, $HOClO_2$

v. Derive van't Hoff general solution equation for 'n' moles of solute.

vi. Represent a cell consisting of $Mg^{2+}$ | Mg half cell and $Ag^{+}$ | Ag half cell and write the cell reaction. $\left(E^{\circ}_{Ag} = 0.799\ V,\ \ E^{\circ}_{Mg} = -2.37\ V\right)$

vii. a. Define the order of chemical reaction.
   b. Find the overall order of the following reactions:
      1. $CHCl_{3(g)} + Cl_{2(g)} \rightarrow CCl_{4(g)} + HCl_{(g)}$, Rate = $k[CHCl_3][Cl_2]$
      2. $2NO_{(g)} + O_{2(g)} \rightarrow 2NO_{2(g)}$, Rate = $k[NO]^2[O_2]$

viii. State and explain Hess's law of constant heat summation.

**Q.2. Attempt any THREE:** **[9]**

i. Explain the following with the help of balanced chemical equation:
   a. Bleaching action of $SO_2$.
   b. Dehydration of formic acid by concentrated $H_2SO_4$.
   c. Burning of benzene in presence of excess of dioxygen.

ii. The density of silver having atomic mass 107.8 g $mol^{-1}$ is 10.8 g $cm^{-3}$. If the edge length of cubic unit cell is $4.05 \times 10^{-8}$ cm, find the number of silver atoms in the unit cell.
   ($N_A = 6.022 \times 10^{23}$, 1 Å = $10^{-8}$ cm)

iii. How much quantity of electricity in coulomb is required to deposit $1.346 \times 10^{-3}$ kg of Ag in 3.5 minutes from $AgNO_3$ solution?
   (Given : Molar mass of Ag is $108 \times 10^{-3}$ kg $mol^{-1}$)

iv. 5 moles of helium expand isothermally and reversibly from a pressure $40 \times 10^{-5}$ N $m^{-2}$ to $4 \times 10^{-5}$ N $m^{-2}$ at 300 K. Calculate the work done, change in internal energy and heat absorbed during the expansion. (R = 8.314 J $K^{-1}$ $mol^{-1}$)

**Q.3. Answer the following:** **[7]**

i. Write structure and molecular formula for the following compounds:
   a. orthophosphoric acid b. sulphurous acid

ii. What is 'Ellingham diagram'? Write any 'two points' of its significance.

iii. In a first order reaction A → product, 80% of the given sample of compound decomposes in 40 min. What is the half life period of the reaction?

**OR**

**Answer the following:**

i. Write features of reversible process.

ii. Write any 'four points' of difference between fluorine and other halogens.

iii. What is freezing point of a liquid? The freezing point of pure benzene is 278.4 K. Calculate the freezing point of the solution when 2.0 g of a solute having molecular weight 100 g $mol^{-1}$ is added to 100 g of benzene. ($K_f$ of benzene = 5.12 K kg $mol^{-1}$.)

**Q.4. Select and write the most appropriate answer from the given alternatives for each sub-question:** **[7]**

i. A substance which shows the highest entropy is _____.

(A) $SrCO_{3(s)}$ (B) $Cu_{(s)}$

(C) $NaCl_{(aq.)}$ (D) $Cl_{2(g)}$

ii. The process of extracting a soluble material from an insoluble solid by dissolving out in a suitable solvent is known as _____.

(A) calcination (B) roasting

(C) leaching (D) smelting

iii. The kinetic order for the following reaction is _____.

$2N_2O_{(g)} \xrightarrow{Pt} 2N_{2(g)} + O_{2(g)}$

(A) zero (B) first

(C) second (D) third

iv. Which of the following compounds of chlorine is used as refrigerant?

(A) $CCl_3NO_2$ (B) $CCl_2F_2$

(C) $COCl_2$ (D) $CCl_4$

v. 4.0 grams of NaOH (Molar mass = 40.0 g $mol^{-1}$) is dissolved in 500 $cm^3$ of water. What is the molarity of NaOH solution?

(A) 1 M (B) 0.8 M

(C) 0.5 M (D) 0.2 M

vi. Number of types of orthorhombic unit cell is _____.

(A) 7 (B) 3 (C) 4 (D) 2

vii. The S.I. unit of cell constant for conductivity cell is _____.

(A) $m^{-1}$ (B) S $m^{-2}$

(C) $cm^{-2}$ (D) S $dm^2$ $mol^{-1}$

## SECTION – II

**Q.5. Attempt any SIX:** **[12]**

i. Explain the geometry of $[Cu(NH_3)_4]^{2+}$ on the basis of hybridisation [At. No. Cu = 29].

ii. What are 'd' and 'f' block elements?

iii. Write a short note on Sandmeyer's reaction.

iv. What is the action of the following on isopropyl methyl ether?

a. Cold HI b. Hot HI

v. Write balanced equations for the following conversions:

a. Cyclopropane carboxylic acid to cyclopropylmethanol.

b. Acetyl chloride to benzyl methyl ketone using dialkyl cadmium.

vi. What is the action of lithium aluminium hydride in the presence of ether on the following compounds?

a. Nitroethane b. 2-Methyl-1-nitropropane.

vii. How is glucose prepared from starch?

viii. Define:

a. Analgesics b. Antimicrobials

**Q.6. Attempt any THREE:** **[9]**

i. Complete and rewrite the balanced chemical equation for the following reactions:

a. 1-chloro-2,4-dinitrobenzene $\xrightarrow[368\ K,\ H^+]{aq.alkali}$ ?

b. Benzaldehyde $\xrightarrow{50\%KOH}$ ?

c. Acetone + phenylhydrazine $\xrightarrow{H^+}$ ?

ii. Write a preparation of phenol from cumene? What happens when phenol is heated with zinc dust?

iii. a. What are racemates?

b. What is the action of following reagents on glucose?

1. Bromine water 2. Hydroxylamine

iv. a. What are elastomers?

b. Distinguish between thermoplastic polymers and thermosetting polymers.

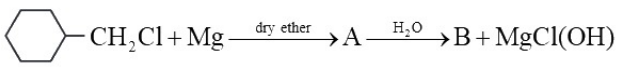
**Q.7. Answer the following:** **[7]**

i. What is the action of the following on lanthanoids?

a. water b. sulphur, heat c. nitrogen, heat

ii. Draw the structures of veronal and thymine.

iii. Identify A and B from the following reaction and rewrite complete reaction:



$$\text{C}_6\text{H}_5\text{CH}_2\text{Cl} + \text{Mg} \xrightarrow{\text{dry ether}} \text{A} \xrightarrow{H_2O} \text{B} + \text{MgCl(OH)}$$

**OR**

**Answer the following:**

i. Write the preparation of benzoic acid from the following:

a. styrene b. benzamide c. dry ice

ii. Write resonance structures of aniline. What is the action of benzene diazonium chloride on ethanol?

iii. Write the formula for pentaamminechlorocobalt (III) sulphate.

**Q.8. Select and write the most appropriate answer from the given alternatives for each sub-question:** **[7]**

i. A polymer which contains ester linkage is_____.

(A) teflon (B) buna-N (C) dextron (D) neoprene

ii. An antihistamine drug is _____.

(A) salvarsan (B) chloramphenicol

(C) seldane (D) prontosil

iii. Identify A and B respectively in the following reaction:

tert-butyl alcohol $\xrightarrow[363K]{20\%H_2SO_4}$ A + $H_2O$ $\xrightarrow[H_2O_2]{HBr}$ B

(A) 2-Methylpropene, 1-bromo-2-methylpropane

(B) 2-Methylpropene, 2-bromo-2-methylpropane

(C) 2-Methylpropane, 1-bromo-2-methylpropane

(D) 2-Methylpropane, 2-bromo-2-methylpropane

iv. The ligand triethylenetetramine is _____.

(A) monodentate (B) bidentate

(C) tridentate (D) tetradentate

v. Pyrolusite ore is _____.

(A) MnO (B) $MnO_2$ (C) $Mn_2O_7$ (D) $Mn_2O_3$

vi. $C_2H_5 - Br + NaI \xrightarrow{\text{dry acetone}} C_2H_5 - I + NaBr$

The above reaction is _____.

(A) Wurtz reaction (B) Balz-Schiemann reaction

(C) Swarts reaction (D) Finkelstein reaction

vii. The functional group present in triacylglycerol is _____.

(A) alcohol (B) ether

(C) ester (D) amine